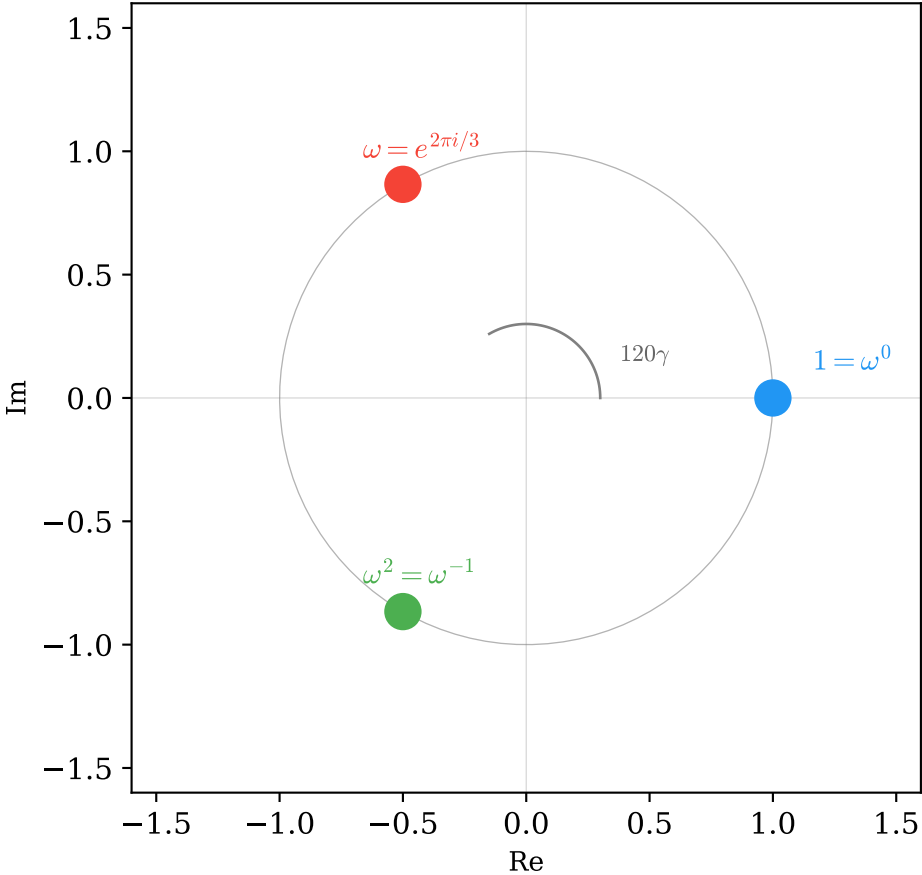
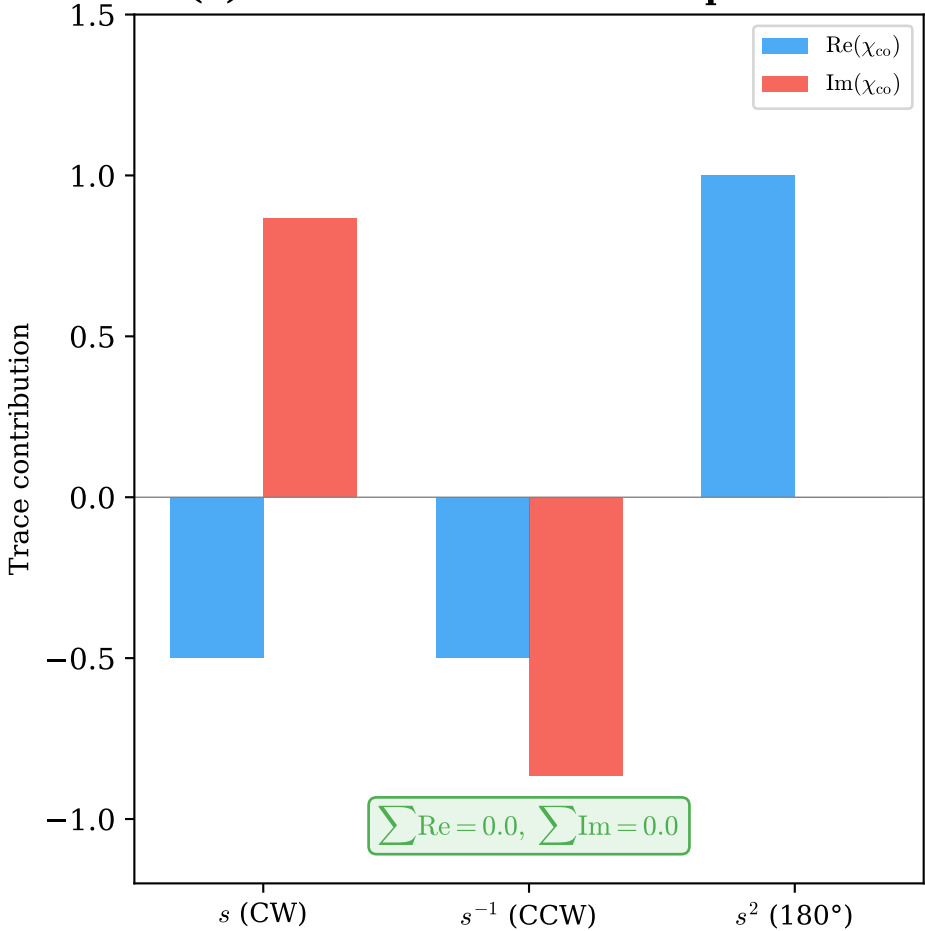


(a) ω^k on the unit circle



(b) Corner-orientation trace per move



(c) Face-sum cancellation

Move	χ_{co}	Re	Im
s (CW)	ω	$\cos(120\gamma)$	$\sin(120\gamma)$
s^{-1} (CCW)	ω^2	$\cos(240\gamma)$	$-\sin(120\gamma)$
s^2 (180°)	1	1	0

$$\omega + \omega^2 + 1 = 0$$
$$\Downarrow$$

Imaginary parts cancel per face

$$\Downarrow$$
$$\chi_{\text{face}} = \chi(s) + \chi(s^{-1}) + \chi(s^2) \in \mathbb{Z}$$
$$\Downarrow$$
$$\lambda \in \mathbb{Q}$$